2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE) | 979-8-3503-7156-7/24/$31.00 ©2024 IEEE | DOI: 10.1109/AMATHE61652.2024.10582142

Implementation of PWM Using RISC-V Processor

Prem Kanjarbhat
Electrical and Electronics Engineering
KLE Technological University
Hubballi, India
premkanjarbhat@gmail.com

Pooja Chandaragi
Electrical and Electronics Engineering
KLE Technological University
Hubballi, India
pooja.appanavar@kletech.ac.in

Uday V Wali
Electrical and Electronics Engineering
KLE Technological University
Hubballi, India
uday.wali@kletech.ac.in

***Abstract*—Implementation of Pulse width modulation(PWM) using RISC-V processor is the main topic of this work. RISC-V is a free and open-source Instruction Set Architecture (ISA) designed for cloud computing, embedded systems, and academic use. To support the development, testing, and optimization of RISC-V-based systems, a diverse ecosystem of emulators and simulators has emerged. This work provides in-depth information on key development tools within the RISC-V ecosystem, aiming to offer valuable insights into the landscape of RISC-V emulators and simulators. The tools, including QEMU, Xilinx Vivado, BRISC-V, WEBRISC-V, Emulsiv, and Visual Studio Code with Gtkwave, support RISC-V software and hardware development. PWM functionality is achieved through a RISC-V processor operating at a frequency of 100MHz, developed using Xilinx Vivado software. The implementation is thoroughly tested through Vivado simulation and validated on an Artix-7 FPGA board.**

***Index Terms*—PWM, RISC-V, FPGA Implementation, Emulators, Simulators.**

I. Introduction

RISC-V's open-source[1, 2] nature has not only democratized access to computer architecture but has also fostered a collaborative environment where developers and researchers actively contribute to its evolution. The open availability of specifications empowers innovators to explore and experiment without the constraints of licensing fees, driving a diverse range of applications[3] and use cases. The benefits of modularization, low power consumption, fast performance, and expandable customization of the open-source RISC-V architecture more than offset the drawbacks of the ARM architecture. It also has no patent constraints.[4]

The modular design of RISC-V plays a pivotal role in its adaptability. Standard extensions, which can be seamlessly added to the base architecture, enable developers to tailor the instruction set to meet specific requirements. This flexibility is particularly advantageous in industries such as cloud computing and embedded systems, where specialized optimizations can significantly enhance performance.

Adhering to a five-stage pipeline architecture is one of the fundamental design concepts that support RISC-V's effectiveness and performance.[5] The execution of an instruction is broken down into five separate steps in a five-stage pipeline: memory access, write-back, fetch, decode, and execution. By enabling the simultaneous execution of several instructions, this pipelining technique improves processor throughput and overall performance.[6]

In the realm of software development and research, the importance of emulators and simulators for RISC-V cannot be overstated. These tools serve as essential aids in various stages of development, from initial coding to testing and experimentation. While simulators focus on achieving functional accuracy, emulators prioritize closely mimicking the performance of the original hardware or software environment.

Developers often face a choice between simulators and emulators based on their specific needs. Simulators may not capture the intricacies of performance but are valuable for functional testing and early-stage development. On the other hand, emulators, by closely replicating the original system's performance characteristics, prove indispensable when optimizing for speed is a primary concern.

The behavior of CPU execution tasks is influenced by several factors, primarily mutual exclusion when sharing resources and synchronous transmissions between jobs that have priority requirements. When it comes to real-time task scheduling, the most crucial concerns that need to be handled are keeping an eye on task deadlines, estimating the likelihood that tasks won't be completed by the deadline, and making sure that the controlled system's deadline is met. This concludes that if the system has sufficient resources to do the tasks without missing any deadlines, then the tasks are viable.[7]

Cloud computing systems are a good fit for RISC-V because of its open-source nature and modular design. Its open ecosystem promotes innovation and developer collaboration, and its flexible instruction set enables the efficient execution of a wide range of workloads. RISC-V is perfect for embedded systems because of its tiny size, low power consumption, and scalability. The instruction set can be customized by developers to match the unique needs of embedded applications because to its modular architecture, which maximizes performance and resource efficiency.

RISC-V is a useful teaching tool for computer architecture and system design because of its open specifications and copious documentation. Students can study a variety of subjects, from processor design to system-level optimization, thanks to its adaptability and simplicity, which also make learning and experimenting easier.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.

The RISC-V ecosystem boasts a plethora of development tools that cater to different aspects of the architecture. Among these tools, QEMU, Xilinx Vivado, BRISC-V, WEBRISC-V, Emulsiv, Virtual Studio Code (VS Code), and Gtkwave stand out for their popularity and functionality. Each tool brings its unique strengths to the table, contributing to the overall vibrancy of the RISC-V development landscape. As the RISC-V ecosystem continues to grow, these tools play a crucial role in advancing the capabilities and applications of this open and adaptable instruction set architecture.

A portion of the research that has been done over the past years on various RISC-V architecture-related topics has been compiled and examined in survey papers that concentrate on the security [8, 9, 10, 11] and open-sourcing [12, 13] elements of the topic. Despite the large number of research that have examined the RISC-V software stack, no report exists in the literature that unifies the analysis of these studies. motivated by the expansion of the RISC-V community and the paucity of research on the condition of this ISA's software at the moment. In [14] RISC-V extension for PWM control is demonstrated. This work focuses on the different emulators and simulators of RISC-V, through a detailed examination of the surveyed tools, this work sheds light on their strengths, limitations, and specific use cases, this work seeks to fill the gap and explore the different emulators and simulators that supports RISC-V architectures. Additionally, this survey highlights the versatile applications of RISC-V emulators across various domains, showcasing their adaptability in diverse fields. By offering insights into evolving trends, challenges, and applications. In this work, PWM is achieved using a RISC-V processor which has an operating frequency of 100MHz. The output of the PWM can be controlled by two inputs Increase Duty and Decrease duty, the output of the PWM is verified in the Vivado simulation and on the Artix-7 FPGA board.

FPGA-based PWM generation's deterministic nature allows for accurate timing control and low latency response, which makes it appropriate for real-time applications where timing accuracy is crucial. FPGA-based PWM can be applied to robotics to improve the responsiveness and performance of robotic systems through motion synchronization, motor control, and sensor data processing. PWM can be used in signal processing applications for digital filtering, high-speed data acquisition, and signal modulation. This allows sensor data and communication signals to be processed in real-time. FPGA-based PWM generators provide flexibility and scalability, allowing developers to adapt the PWM waveform to meet the specific requirements of their applications. This enables fine-grained control over motor speed, voltage output, and signal modulation, resulting in improved performance and efficiency. In FPGA-based PWM generators users can fine-tune the PWM waveform to meet their unique needs by using features that are responsive and simple to use, like push buttons for changing duty cycle and frequency.

II. Various development tools of RISC-V

The survey explores various development tools for the RISC-V architecture, which is an open-source instruction set architecture (ISA). The tools discussed include:

- QEMU
- Xilinx Vivado
- BRISC-V
- WEBRISC-V
- Emulsiv
- Visual Studio Code(VS Code) and Gtkwave

A. QEMU

- QEMU stands for Quick Emulator.
- It is a popular open-source emulator that supports various architectures, including RISC-V [15].
- QEMU allows developers to run RISC-V programs on different host platforms for testing and development purposes.
- It is highly configurable and widely used in the RISC-V ecosystem for software development and debugging.

B. Xilinx Vivado

- Xilinx Vivado is a commercial integrated development environment (IDE) for FPGA (Field-Programmable Gate Array) and SoC (System-on-Chip) design.
- Vivado supports the integration of RISC-V cores into FPGA and SoC designs.
- It provides a comprehensive suite of tools for hardware development, including synthesis, simulation, and implementation.

C. BRISC-V

- BRISC-V, or Berkeley RISC-V, is an open-source work that aims to provide a highly configurable and extensible RISC-V processor generator.
- It enables researchers and developers to customize RISC-V cores to meet specific application requirements [16].
- BRISC-V provides a framework for generating RISC-V processor designs with various features and optimizations [17].

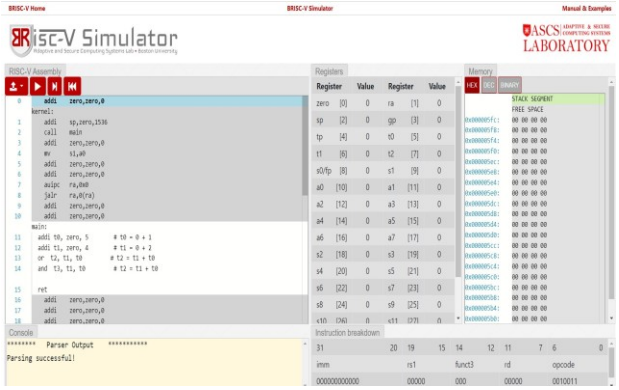


Fig 1. BRISC-V

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.

D. WEBRISC-V

- WEBRISC-V is an online platform that offers a web-based integrated development environment for RISC-V [18].
- It provides a convenient and accessible way for developers to write, compile, and run RISC-V programs directly in a web browser.
- WEBRISC-V is a user-friendly tool, especially for educational purposes and quick prototyping.

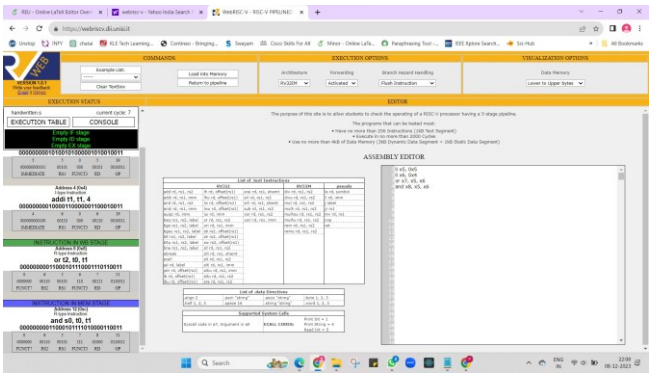


Figure 2. WEBRISC-V

E. Emulsiv

- Emulsiv is an open-source RISC-V simulator designed for research and development purposes.
- It offers a flexible and extensible architecture, allowing users to explore different RISC-V configurations and extensions.
- Emulsiv is particularly useful for exploring novel RISC-V designs and conducting experiments in the RISC-V ecosystem.
- This tools play crucial roles in the development, testing, and exploration of RISC-V-based systems, offering a range of options for both hardware and software development in the RISC-V architecture.

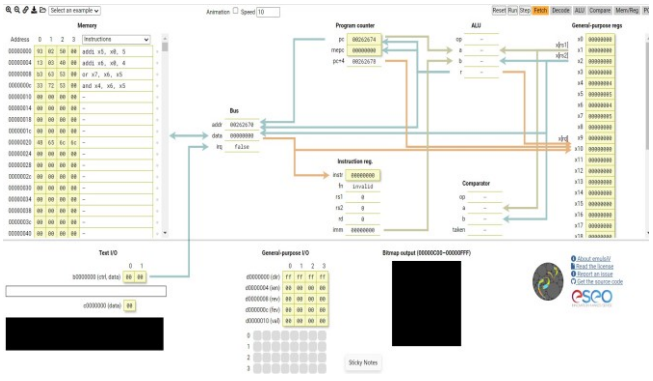


Fig 3. EMULSIV

F. Visual Studio Code(VS Code) and Gtkwave

- Visual Studio Code is a popular open-source code editor that provides a versatile environment for software development and coding.
- Gtkwave is an open-source waveform viewer used for viewing simulation results and timing diagrams in digital design and verification.
- Together, VS Code and Gtkwave offer comprehensive support for RISC-V development, combining a powerful code editing environment with the capability to visualize and analyze simulation results, making them valuable tools for both software and hardware development in the RISC-V ecosystem.

The comparison of RISC-V tools based on type, supported ISA levels and applications is given in Table I.

III. Methodology

In this work, Xilinx Vivado is utilized for implementing Pulse Width Modulation (PWM) using a RISC-V processor. Leveraging the comprehensive features of Vivado, including synthesis, simulation, and implementation, facilitates the seamless integration of RISC-V cores into FPGA designs. One of the notable advantages of utilizing Vivado is the ability to visualize and validate the output of the PWM implementation directly on an FPGA board. This enables real-time testing and verification, providing invaluable insights into the functionality and performance of the RISC-V based PWM system. By harnessing the power of Vivado, we ensure the robustness and reliability of our implementation, setting a solid foundation for further experimentation and optimization.

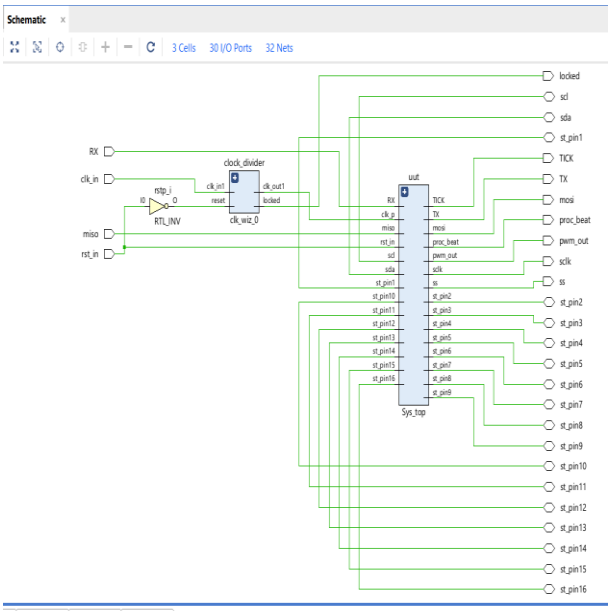


Fig 4. Schematic Diagram in Vivado

The implemented design represents a PWM (Pulse Width Modulation) generator system featuring essential components

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.

Table I. Comparison of RISC-V Tools

Tool	Type	Supported ISA Levels	Applications
QEMU	Emulator	RV32I, RV64I	Software Development, Testing
Xilinx Vivado	IDE/Toolchain	RV32I, RV64I	FPGA Design, Prototyping
BRISC-V	Simulator	RV32I, RV64I	Education, Research
WEBRISC-V	Web-based	RV32I	Learning RISC-V Architecture
Emulsiv	Simulator	RV32I, RV64I	Education, Research
VS Code and Gtkwave	Code development, and Visualization	RV32I, RV64I	Code development, Simulation result visualization

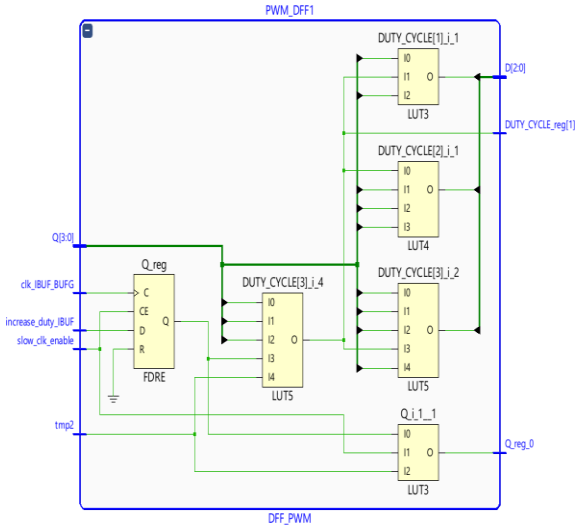


Figure 5. Block Diagram of PWM

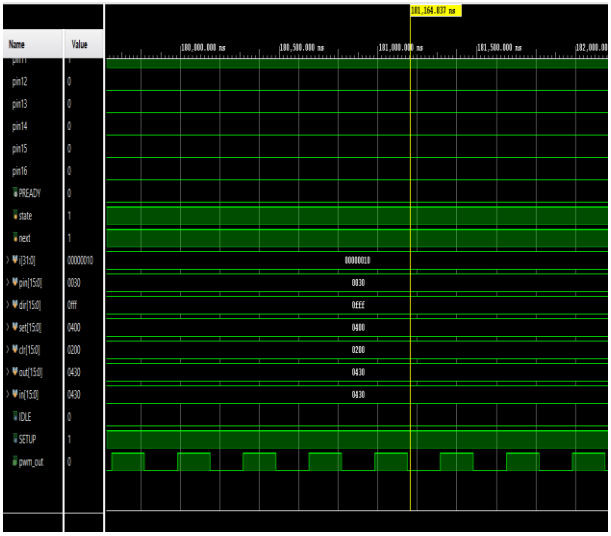


Figure 6. Simulation Output in Vivado

such as clock input, increase duty, decrease duty, and PWM output. These signals collectively form a versatile platform for dynamically adjusting the duty cycle of the output pulse waveform, crucial in various applications requiring precise control over power delivery and signal modulation. The increase duty and decrease duty inputs serve as intuitive control signals, enabling users to adjust the width of the output pulse with ease. Figures 4 and Figure 5 illustrate the schematic diagram of the system, depicting the interconnection of components and signal paths. The modular design of the system facilitates seamless integration into larger digital systems or FPGA-based applications, offering flexibility and adaptability to meet diverse project requirements. The implemented PWM is tested on Artix-7 FPGA board.

IV. Results and Discussion

Figure 6 shows the simulation output obtained in Vivado. The waveform showcases a pulse signal characterized by a 50% duty cycle, indicating that the pulse is active for halfof the total period. Additionally, the magnitude of the output pulse is observed to be 50% of the corresponding input signal.

The output of the PWM is verified on the Artix-7 FPGA board. The increase in duty cycle is assigned to push button BTN0, while the decrease in duty cycle is assigned to push button BTN1. The PWM output is then assigned to RGBLED LD4 on the FPGA board. Initially, the duty cycle is set to 0%, resulting in the PWM output being zero and the LED being OFF, as shown in Figure 6. When BTN0 is pressed, the duty cycle increases, widening the output pulse and turning the LED ON. Duty cycle can be increased or decreased according to the requirements from 0-100%. As shown in Figure 8, the duty cycle is set to 50% by pressing the increase duty button BTN0, resulting in the LED being 50% illuminated. Similarly, in Figure 9, the duty cycle is set to 100% by pressing BTN0, causing the LED to be fully illuminated. The duty cycle can be decreased by pressing the decrease duty push button BTN1. By adjusting the duty cycle through the push buttons BTN0 and BTN1, users can finely control the output of PWM, allowing for precise adjustments in various applications. This setup demonstrates the versatility of using FPGA-based PWM generation not only for controlling LEDs but also for driving other hardware components such as motors. With this integration, the FPGA becomes a powerful tool for real-time control and automation tasks in diverse fields ranging from robotics to industrial automation.

A RISC-V CPU running at 100MHz can be used to implement PWM capability. This setup provides several performance advantages, such as low latency, high precision, and effective resource use. With very little jitter and distortion, the PWM waveform produced by the RISC-V processor be- haves steadily and consistently. Applications needing precise

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.

timing and modulation can benefit from this as it guarantees accurate control of output signals. The performance of PWM capability can be assessed by taking into account efficiency criteria including duty cycle accuracy, frequency stability, and response time. Developers can attain maximum efficiency and dependability in their applications by refining these measurement systems.

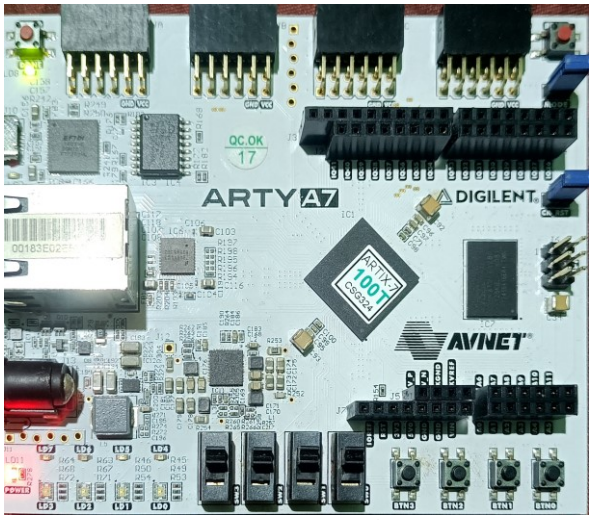


Figure 7. Duty Cycle=0

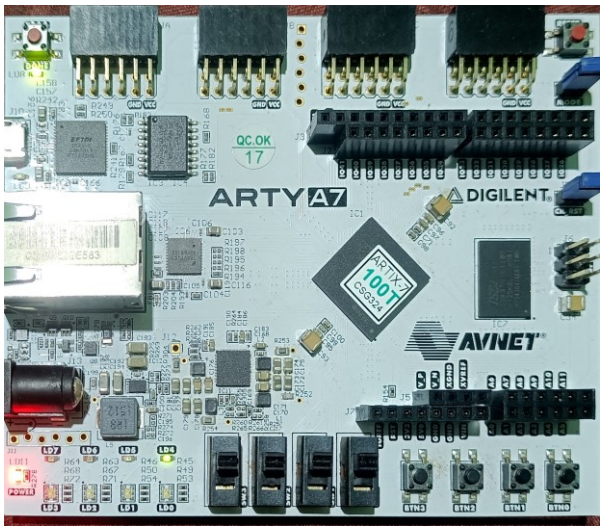


Figure 8. Duty Cycle=50%

V. Conclusion

In this work, implementation of Pulse Width Modulation (PWM) using RISC-V processor is presented. Provided control inputs for adjusting the duty cycle of the output waveform. Through simulations in Vivado and validation on an Artix-7 FPGA board, demonstrated the functionality and versatility of the implemented PWM system. The user-friendly interface

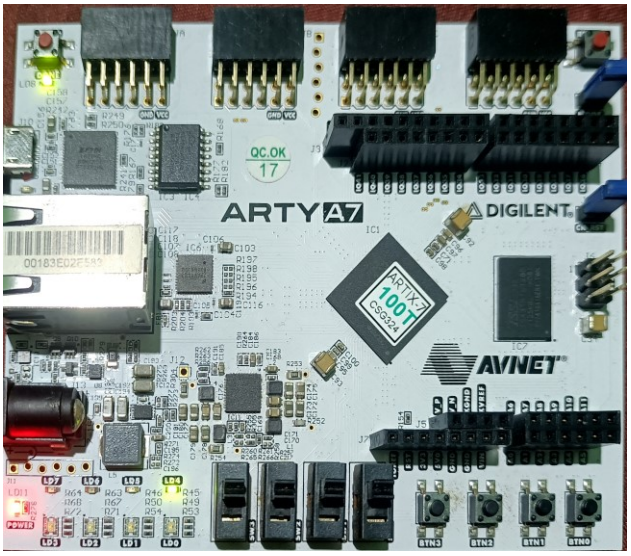


Figure 9. Duty Cycle=100%

enhances usability, allowing for the adjustment of PWM signal characteristics based on specific application needs. The presence of dedicated inputs for duty cycle adjustment simplifies user interaction, while the potential for future enhancements ensures the system's relevance and applicability in evolving technological landscapes.

The integration of FPGA-based PWM generation offers significant advantages in various domains, including motor control, power regulation, and signal processing. With this integration, the FPGA becomes a powerful tool for real-time control and automation tasks, with implications for robotics, industrial automation, and beyond. Overall, the implemented PWM generator system represents a significant advancement in signal modulation and power control. Its modular design and adaptability make it suitable for a wide range of applications, from educational purposes to industrial automation.

VI. Future Scope

The implementation of Pulse Width Modulation (PWM) using RISC-V processors and FPGA-based technology presents a promising avenue for future research and development. Future research avenues and enhancements can significantly expand the scope of this work. Potential areas for exploration include delving into advanced modulation techniques, optimizing performance metrics such as latency and precision, studying real-time applications requiring stringent timing constraints, and fostering open-source contributions to the RISC-V ecosystem. Specific enhancements could focus on refining PWM techniques, integrating the system with emerging technologies or communication protocols, optimizing energy efficiency, and improving user interface and interaction capabilities. Scalability in the face of evolving technological landscapes can be ensured through adaptable designs, efficient resource utilization, and robust compatibility with future hardware and software advancements.

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.

VII. ACKNOWLEDGEMENT

The authors would like to acknowledge the support extended by the authorities of KLE Technological University, Hubballi, 580031, India in carrying out this work.

REFERENCES

[1] Waterman, A.; Asanović, K. (Eds.) The RISC-V Instruction Set Manual; Volume I: Unprivileged ISA; University of California: Berkeley, CA, USA, 2019.

[2] Waterman, A.; Asanović, K. (Eds.) The RISC-V Instruction Set Manual; Volume II: Privileged Architecture; University of California: Berkeley, CA, USA, 2021.

[3] G. S. Nicholas, Y. Gui and F. Saqib, "A Survey and Analysis on SoC Platform Security in ARM, Intel and RISC-V Architecture," 2020 IEEE 63rd International Midwest Symposium on Circuits and Systems (MWSCAS), Springfield, MA, USA, 2020, pp. 718-721, doi: 10.1109/MWSCAS48704.2020.9184573.

[4] J. Gao and J. Zhang, "Research and Design of RISC-V Four-Stage Out-of-Order Execution Processor," 2022 IEEE 16th International Conference on Solid-State Integrated Circuit Technology (ICSICT), Nangjing, China, 2022, pp. 1-3, doi: 10.1109/ICSICT55466.2022.9963329.

[5] M. Kalmath, A. Kulkarni, S. V. Siddamal, and J. Mallidu, "Implementation of 32-bit isa five-stage pipeline risc-v processor core," in Artificial Intelligence and Sustainable Computing, M. Pandit, M. K. Gaur, P. S. Rana, and A. Tiwari, Eds. Singapore: Springer Nature Singapore, 2022, pp. 243–251.

[6] J. Mallidu and S. V. Siddamal, "A survey on in-order 5-stage pipeline risc-v processor implementation," in Information and Communication Technology for Competitive Strategies (ICTCS 2021), M. S. Kaiser, J. Xie, and V. S. Rathore, Eds. Singapore: Springer Nature Singapore, 2023, pp. 777–784

[7] N. Iuga, I. Zagan and V. G. Gaitan, "CPU Execution Time Analysis based on RISC-V ISA Simulators: A Survey," 2022 International Conference on Development and Application Systems (DAS), Suceava, Romania, 2022, pp. 12-18, doi: 10.1109/DAS54948.2022.9786163.

[8] G. S. Nicholas, Y. Gui and F. Saqib, "A survey and analysis on SoC platform security in ARM Intel and RISC-V architecture", Proc. IEEE 63rd Int. Midwest Symp. Circuits Syst. (MWSCAS), pp. 718-721, Aug. 2020.

[9] T. Lu, "A survey on RISC-V security: Hardware and architecture", arXiv:2107.04175, 2021.

[10] R. Newell, J. Xie and H. Handschuh, "Survey of notable security-enhancing activities in the RISC-V universe", Proc. 17th Int. Workshop Cryptograph. Architectures Embedded Log. Devices (CryptArchi), pp. 1-10, 2019,

[11] B. Marshall, G. R. Newell, D. Page, M.-J.-O. Saarinen and C. Wolf, "The design of scalar AES instruction set extensions for RISC-V", IACR Trans. Cryptograph. Hardw. Embedded Syst., vol. 2021, no. 1, pp. 109-136, Dec. 2020.

[12] A. Dörflinger, M. Albers, B. Kleinbeck, Y. Guan, H. Michalik, R. Klink, et al., "A comparative survey of open-source application-class RISC-V processor implementations", Proc. 18th ACM Int. Conf. Comput. Frontiers, pp. 12-20, May 2021.

[13] I. Elsadek and E. Y. Tawfik, "RISC-V resource-constrained cores: A survey and energy comparison", Proc. 19th IEEE Int. New Circuits Syst. Conf. (NEWCAS), pp. 1-5, Jun. 2021.

[14] C. -A. Popovici, A. Stan and V. -I. Manta, "RISC-V Extension for Optimized PWM Control," 2023 27th International Conference on System Theory, Control and Computing (ICSTCC), Timisoara, Romania, 2023, pp. 404-409,

[15] Fabrice Bellard "QEMU, a Fast and Portable Dynamic Translator" FREENIX Track: 2005 USENIX Annual Technical Conference

[16] Mihailo Isakov and Michel A. Kinsy "BRISC-V Emulator: A Standalone, Installation-Free, Browser-Based Teaching Tool" arXiv:1812.04077v1 [cs.PL] 7 Dec 2018

[17] Sahan Bandara, Alan Ehret, Donato Kava and Michel A. Kinsy "BRISC-V: An Open-Source Architecture Design Space Exploration Toolbox" arXiv:1908.09992v1 [cs.AR] 27 Aug 2019

[18] Mariotti, Gianfranco, Giorgi, Roberto " WebRISC-V: A RISC-V Educational Simulator featuring RV64IM, Pipeline and Web-Based UI," PY - 2020/07/15

Authorized licensed use limited to: UNIVERSITAS GADJAH MADA. Downloaded on August 28,2026 at 06:15:42 UTC from IEEE Xplore. Restrictions apply.